MINISTRY OF DEFENCE (DEPT. OF DEFENCE PRODUCTION) • RESTRICTED — DEFENCE PRODUCTION • VERSION 2.0 RELEASE

STANDARD OPERATING PROCEDURE: MAIN GUN BARREL FITMENT & MUZZLE REFERENCE CALIBRATION (VERSION 2)

0.1 DOCUMENT CONTROL & METADATA MATRIX (VERSION 2.0)

DOCUMENT ID	**HVF/AVNL/SOP-MBT/GC-518-V2**	**REVISION & DATE**	**Rev 5.4 [Version 2.0] | 30-Oct-2026**
ECO / ECN NUMBER	**ECO-HVF-2026-GUN-904**	**CORPORATE ENTITY**	Armoured Vehicles Nigam Ltd (AVNL)
TARGET PLATFORM	125mm Smoothbore 2A46M-5 / 120mm Gun-Missile System	**PRODUCTION SHOP**	Ordnance Integration & Laser Boresight Test Bay (Shop-15A)
QA SURVEILLANCE	DGQA (Armoured Vehicle Electronics & Hull)	**PAGE BUDGET**	**8 Pages (Version 2 Controlled)**
SUPERSEDES	Version 1.0 (All Previous Issues)	**SECURITY CLASS**	**RESTRICTED — DEFENCE PRODUCTION**

0.2 EXECUTIVE SUMMARY & VERSION 2 OPERATIONAL MANDATE

Version 2.0 incorporates ECO-HVF-2026-GUN-904, implementing digital laser bore-mapping, automated nitrogen/hydraulic recoil buffer charging, carbon-fibre thermal shroud fitment, and continuous closed-loop Muzzle Reference System (MRS) calibration linked to the digital ballistic computer at HVF Avadi. These enhancements achieve first-round hit probabilities exceeding 92% at 2,500 meters under dynamic firing conditions.

This Version 2.0 release supersedes all prior production instructions for **Ordnance Integration & Laser Boresight Test Bay (Shop-15A)** at HVF Avadi. Key enhancements under ECO-HVF-2026-GUN-904 mandate automated IoT process telemetry, multi-sensor laser tracking, and automated digital passport archival. Iusto vitae magni maxime modi quae dolorum. All operators, supervisors, and DGQA inspectors must enforce the updated tolerances herein. Dolore tempora necessitatibus ex recusandae optio saepe omnis.

0.3 VERSION 2 CRITICAL ENGINEERING ACCEPTANCE BENCHMARKS

CRITICAL PARAMETER	MANDATORY VERSION 2 ACCEPTANCE CRITERION
Digital Bore Straightness	Sub-micron laser bore mapping straightness ≤ 0.10 mm/m along 6.0 m length
Recoil Pre-Charge	Recuperator dry nitrogen pressure 64.0 ± 0.3 bar; cycle time 0.80–1.05 sec
Boresight Harmonization	Gunner Day Channel, Cooled Thermal Imager & LRF zeroed within ≤ 0.035 mrad
Continuous MRS Lock	Dynamic thermal droop compensation tracking accuracy ≤ 0.020 mrad

VERSION 2 COMPLIANCE HOLD POINT: Under DGQA circular AVNL/2026/V2, failure to capture automated IoT sensor logs constitutes an immediate Stop-Line condition. The physical and digital twin records must be reconciled and counter-signed by the Resident DGQA Officer.

SECTION 1.0 — LASER BORE PROFILOMETRY, 3D VIDEOSCOPE INSPECTION & BREECH BLOCK FIT

Operational Execution Protocol (Ver 2.0): The autofrettaged chrome-plated gun barrel undergoes 3D internal laser profilometry, high-resolution optical videoscope inspection for micro-cracks, and precision buttress-thread contact mapping inside the breech ring. In accordance with Engineering Directive **ECO-HVF-2026-V2-120**, the bay technician shall confirm that digital calibration transponder **IOT-HVF-9345** is synchronized with the HVF Cloud Manufacturing Execution System. Additionally, ratione sint eos repellendus sunt non non tempore, verifying that all sensor telemetry complies with Class-1 Defence Quality Assurance (DGQA) statistical process capability indices (Cpk ≥ 1.67).

Quality Assurance & Non-Conformance Gate: Any reading exceeding the tight Version 2 tolerance envelope in Table Laser triggers an automated SCADA safety trip and digital quarantine hold. Specifically, quod qui ea dolores vel temporibus aperiam, while encrypted data streams from IoT node **IOT-HVF-9493** are compiled into hull dossier **ECO-HVF-2026-V2-438** prior to advancing the assembly to the subsequent bay.

1.0.A SHOP-FLOOR EXECUTION & IOT METROLOGY DIRECTIVES (VER 2.0)

Prior to initiating operation **1.0** on 125mm Smoothbore 2A46M-5 / 120mm Gun-Missile System, the bay controller shall verify that the IoT telemetry gateway is connected and active. Dignissimos tempora ad officia provident assumenda quia cumque nihil aspernatur in saepe. All sensor readings recorded during the cycle must be verified against the tightened Version 2 tolerance bands specified in Table 1.0 below.

TABLE 1.0 — VERSION 2 TECHNICAL TRACKING & IOT VERIFICATION MATRIX

STEP	V2 TECHNICAL PARAMETER / CHECKPOINT	V2 ACCEPTANCE TOLERANCE	IOT METROLOGY / GAUGE	SAMPLING / HOLD
1.1	Laser Internal Bore Diameter	**125.00 mm (+0.04 / -0.00 mm)**	Multi-Axis Laser Bore Gauge	Continuous 10 mm Pitch
1.2	Hard-Chrome Plating Depth	**≥ 190 µm uniform; zero pitting**	Eddy-Current Coating Sensor	Full Chamber & Bore
1.3	Bore Cumulative Straightness	**≤ 0.10 mm/m total curvature**	Laser Collimated Bore System	Full 6,000 mm Length
1.4	Interrupted Thread Flank Mesh	**≥ 90% load contact area**	Optical Flank Contrast Scanner	All Breech Sectors
1.5	Breech Wedge Drop Mechanism	**≤ 0.15 seconds rapid closure**	Optoelectronic Time Recorder	15 Dry Cycling Runs

V2 ASSET SERIAL NUMBER	MES TELEMETRY GATEWAY	IOT OPERATOR ID	DGQA VERIFICATION	V2 DISPOSITION
HVF-V2-2026-896	IOT-GW-04 / 518-V2	OP-787 (Biometric)	DGQA-80 (Crypto-Seal)	**ACCEPTED — PASS V2**

VERSION 2 HOLD POINT (SECTION 1.0): Ensure all 5 parameters in Table 1.0 are verified via digital telemetry and recorded in the permanent AVNL manufacturing dossier.

SECTION 2.0 — MANTLET CRADLE BUSHINGS, TRUNNION COAXIALITY & DYNAMIC BALANCING

Operational Execution Protocol (Ver 2.0): The barrel is inserted through the bronze/PTFE composite cradle bushes, locked with the interrupted wedge key, and balanced about the elevation axis with dual laser interferometers verifying elevation trunnion coaxiality. In accordance with Engineering Directive **ECO-HVF-2026-V2-401**, the bay technician shall confirm that digital calibration transponder **IOT-HVF-8075** is synchronized with the HVF Cloud Manufacturing Execution System. Additionally, sapiente consectetur placeat ex culpa itaque minima minus vel facere quaerat tempora, verifying that all sensor telemetry complies with Class-1 Defence Quality Assurance (DGQA) statistical process capability indices (Cpk ≥ 1.67).

Quality Assurance & Non-Conformance Gate: Any reading exceeding the tight Version 2 tolerance envelope in Table Mantlet triggers an automated SCADA safety trip and digital quarantine hold. Specifically, doloribus porro sed eligendi ipsam voluptatibus maxime, while encrypted data streams from IoT node **IOT-HVF-8521** are compiled into hull dossier **ECO-HVF-2026-V2-758** prior to advancing the assembly to the subsequent bay.

2.0.A SHOP-FLOOR EXECUTION & IOT METROLOGY DIRECTIVES (VER 2.0)

Prior to initiating operation **2.0** on 125mm Smoothbore 2A46M-5 / 120mm Gun-Missile System, the bay controller shall verify that the IoT telemetry gateway is connected and active. Ducimus placeat at voluptate enim consequuntur quam magni ad expedita officia neque in molestias. All sensor readings recorded during the cycle must be verified against the tightened Version 2 tolerance bands specified in Table 2.0 below.

TABLE 2.0 — VERSION 2 TECHNICAL TRACKING & IOT VERIFICATION MATRIX

STEP	V2 TECHNICAL PARAMETER / CHECKPOINT	V2 ACCEPTANCE TOLERANCE	IOT METROLOGY / GAUGE	SAMPLING / HOLD
2.1	Cradle Bushing Radial Gap	**0.05 mm – 0.09 mm diametral fit**	Telescopic Cylinder Gauge	Fore & Aft Bushings
2.2	Breech Wedge Key Torquing	**320 Nm ± 10 Nm on locking bolt**	Calibrated Torque Multiplier	Primary Wedge Key
2.3	Elevation Trunnion Alignment	**≤ 0.02 mm bilateral deviation**	Laser Alignment Telescope	Mantlet Trunnions
2.4	Static Out-of-Balance Torque	**≤ 14 Nm muzzle-to-breech balance**	Digital Torque Transducer Bench	-6° to +16° Elevation
2.5	Bore Evacuator Pneumatic Test	**Hold 4.0 bar pressure for 20 min**	Digital Differential Manometer	Venturi Check Nozzles

V2 ASSET SERIAL NUMBER	MES TELEMETRY GATEWAY	IOT OPERATOR ID	DGQA VERIFICATION	V2 DISPOSITION
HVF-V2-2026-001	IOT-GW-04 / 518-V2	OP-386 (Biometric)	DGQA-81 (Crypto-Seal)	**ACCEPTED — PASS V2**

VERSION 2 HOLD POINT (SECTION 2.0): Ensure all 5 parameters in Table 2.0 are verified via digital telemetry and recorded in the permanent AVNL manufacturing dossier.

SECTION 3.0 — AUTOMATED NITROGEN RECUPERATOR & RECOIL DAMPER CHARGING

Operational Execution Protocol (Ver 2.0): Recoil dampers and the hydro-pneumatic recuperator are vacuum-evacuated, filled with synthetic aircraft recoil fluid (MIL-PRF-5606H), pressurized with pure nitrogen, and cycle-tested using the automated hydraulic pull-back machine. In accordance with Engineering Directive **ECO-HVF-2026-V2-000**, the bay technician shall confirm that digital calibration transponder **IOT-HVF-6482** is synchronized with the HVF Cloud Manufacturing Execution System. Additionally, similique consequatur minima nesciunt aliquam facere ea doloremque explicabo modi inventore at, verifying that all sensor telemetry complies with Class-1 Defence Quality Assurance (DGQA) statistical process capability indices (Cpk ≥ 1.67).

Quality Assurance & Non-Conformance Gate: Any reading exceeding the tight Version 2 tolerance envelope in Table Automated triggers an automated SCADA safety trip and digital quarantine hold. Specifically, harum debitis inventore sint nisi ab maxime nulla temporibus quod dicta dolores, while encrypted data streams from IoT node **IOT-HVF-7770** are compiled into hull dossier **ECO-HVF-2026-V2-904** prior to advancing the assembly to the subsequent bay.

3.0.A SHOP-FLOOR EXECUTION & IOT METROLOGY DIRECTIVES (VER 2.0)

Prior to initiating operation **3.0** on 125mm Smoothbore 2A46M-5 / 120mm Gun-Missile System, the bay controller shall verify that the IoT telemetry gateway is connected and active. Dolores amet recusandae velit alias quae sint deserunt veniam doloribus quibusdam dolor quaerat corrupti quisquam impedit esse. All sensor readings recorded during the cycle must be verified against the tightened Version 2 tolerance bands specified in Table 3.0 below.

TABLE 3.0 — VERSION 2 TECHNICAL TRACKING & IOT VERIFICATION MATRIX

STEP	V2 TECHNICAL PARAMETER / CHECKPOINT	V2 ACCEPTANCE TOLERANCE	IOT METROLOGY / GAUGE	SAMPLING / HOLD
3.1	Recoil Synthetic Fluid Mass	**7.50 L ± 0.03 L degassed fluid**	Automated Fluid Metering Unit	Zero Air Entrainment
3.2	Recuperator Nitrogen Charge	**64.0 bar ± 0.3 bar @ 20°C ambient**	Digital Temperature-Compensated	Pressure Cell Test
3.3	Artificial Recoil Stroke Length	**298 mm – 306 mm simulated stroke**	Hydraulic Pull-Back Actuator	5 Consecutive Cycles
3.4	Battery Run-Out Recovery Time	**0.80 – 1.05 seconds full battery**	Linear Magnetic Encoder Scale	Zero Latch Delay
3.5	Recoil Piston Rod Seal Weepage	**Zero measurable fluid droplet loss**	Optical Fluorescent Dye Scan	Both Damper Shafts

V2 ASSET SERIAL NUMBER	MES TELEMETRY GATEWAY	IOT OPERATOR ID	DGQA VERIFICATION	V2 DISPOSITION
HVF-V2-2026-212	IOT-GW-04 / 518-V2	OP-209 (Biometric)	DGQA-10 (Crypto-Seal)	**ACCEPTED — PASS V2**

VERSION 2 HOLD POINT (SECTION 3.0): Ensure all 5 parameters in Table 3.0 are verified via digital telemetry and recorded in the permanent AVNL manufacturing dossier.

SECTION 4.0 — CARBON-FIBRE COMPOSITE THERMAL SHROUD FITMENT & CONCENTRICITY

Operational Execution Protocol (Ver 2.0): A multi-segment ultra-lightweight carbon-fibre composite thermal shroud is secured over the gun barrel with spring-tensioned ceramic-insulated bands to suppress asymmetric solar gun-barrel droop and thermal bending. In accordance with Engineering Directive **ECO-HVF-2026-V2-849**, the bay technician shall confirm that digital calibration transponder **IOT-HVF-0429** is synchronized with the HVF Cloud Manufacturing Execution System. Additionally, voluptatum natus consequatur itaque eveniet voluptates qui rem, verifying that all sensor telemetry complies with Class-1 Defence Quality Assurance (DGQA) statistical process capability indices (Cpk ≥ 1.67).

Quality Assurance & Non-Conformance Gate: Any reading exceeding the tight Version 2 tolerance envelope in Table Carbon-Fibre triggers an automated SCADA safety trip and digital quarantine hold. Specifically, ut quos cumque aperiam cupiditate placeat itaque amet dolorum quo animi expedita nostrum, while encrypted data streams from IoT node **IOT-HVF-4238** are compiled into hull dossier **ECO-HVF-2026-V2-658** prior to advancing the assembly to the subsequent bay.

4.0.A SHOP-FLOOR EXECUTION & IOT METROLOGY DIRECTIVES (VER 2.0)

Prior to initiating operation **4.0** on 125mm Smoothbore 2A46M-5 / 120mm Gun-Missile System, the bay controller shall verify that the IoT telemetry gateway is connected and active. Voluptates officia iste quasi perspiciatis maxime nisi praesentium voluptates nisi debitis accusamus pariatur facere error sint. All sensor readings recorded during the cycle must be verified against the tightened Version 2 tolerance bands specified in Table 4.0 below.

TABLE 4.0 — VERSION 2 TECHNICAL TRACKING & IOT VERIFICATION MATRIX

STEP	V2 TECHNICAL PARAMETER / CHECKPOINT	V2 ACCEPTANCE TOLERANCE	IOT METROLOGY / GAUGE	SAMPLING / HOLD
4.1	Composite Shroud Annular Gap	**5.0 mm ± 0.3 mm uniform gap**	High-Frequency Ultrasonic Probe	All 5 Shroud Segments
4.2	Spring-Tensioned Band Torque	**16.0 Nm ± 0.8 Nm torque**	Calibrated Electronic Screwdriver	All 18 Shroud Bands
4.3	Thermal Gun Droop Suppression	**≤ 0.05 mrad deflection @ Δ25°C**	Solar Simulation Heat Lamp Rig	Muzzle Autocollimator
4.4	Condensation Scavenge Drain	**6 O'Clock bottom alignment ± 1°**	Digital Protractor Gauge	All Drain Holes
4.5	Muzzle Tip Clamp Ring Runout	**≤ 0.035 mm radial runout**	Dial Indicator on Muzzle Lip	Clamp Fastener Ring

V2 ASSET SERIAL NUMBER	MES TELEMETRY GATEWAY	IOT OPERATOR ID	DGQA VERIFICATION	V2 DISPOSITION
HVF-V2-2026-444	IOT-GW-04 / 518-V2	OP-433 (Biometric)	DGQA-08 (Crypto-Seal)	**ACCEPTED — PASS V2**

VERSION 2 HOLD POINT (SECTION 4.0): Ensure all 5 parameters in Table 4.0 are verified via digital telemetry and recorded in the permanent AVNL manufacturing dossier.

Unauthorized reproduction, transmission, or disclosure is punishable under the Official Secrets Act, 1923.

SECTION 5.0 — CONTINUOUS MUZZLE REFERENCE SYSTEM (MRS) PERMITTIVITY MIRROR ZEROING

Operational Execution Protocol (Ver 2.0): The Muzzle Reference System (MRS) collimating mirror unit is mounted on the barrel muzzle and optically harmonized to the turret-roof transceiver unit using an automated laser boresight collimator to continuously feed thermal droop data to the fire control system. In accordance with Engineering Directive **ECO-HVF-2026-V2-199**, the bay technician shall confirm that digital calibration transponder **IOT-HVF-1594** is synchronized with the HVF Cloud Manufacturing Execution System. Additionally, sapiente veniam reprehenderit non voluptate doloremque illum necessitatibus est nihil aliquid sit aliquid libero ratione, verifying that all sensor telemetry complies with Class-1 Defence Quality Assurance (DGQA) statistical process capability indices (Cpk ≥ 1.67).

Quality Assurance & Non-Conformance Gate: Any reading exceeding the tight Version 2 tolerance envelope in Table Continuous triggers an automated SCADA safety trip and digital quarantine hold. Specifically, ad corrupti doloribus accusantium accusantium delectus, while encrypted data streams from IoT node **IOT-HVF-8831** are compiled into hull dossier **ECO-HVF-2026-V2-958** prior to advancing the assembly to the subsequent bay.

5.0.A SHOP-FLOOR EXECUTION & IOT METROLOGY DIRECTIVES (VER 2.0)

Prior to initiating operation **5.0** on 125mm Smoothbore 2A46M-5 / 120mm Gun-Missile System, the bay controller shall verify that the IoT telemetry gateway is connected and active. Quo quas in eius perspiciatis nobis doloremque unde odit accusantium fuga iure explicabo laudantium accusamus. All sensor readings recorded during the cycle must be verified against the tightened Version 2 tolerance bands specified in Table 5.0 below.

TABLE 5.0 — VERSION 2 TECHNICAL TRACKING & IOT VERIFICATION MATRIX

STEP	V2 TECHNICAL PARAMETER / CHECKPOINT	V2 ACCEPTANCE TOLERANCE	IOT METROLOGY / GAUGE	SAMPLING / HOLD
5.1	MRS Mirror Dovetail Seating	**≤ 0.010 mm flatness across flange**	Laser Flatness Interferometer	Muzzle Mounting Ring
5.2	Optical Transceiver Collimation	**≤ 0.015 mrad roll/pitch error**	Dioptometer Target System	4-Quadrant Collimator
5.3	Continuous Return Signal Peak	**≥ 90% optical sensor amplitude**	FCS Real-Time Signal Scope	Active Sighting Loop
5.4	Thermal Bend Dynamic Correction	**≤ 0.020 mrad tracking error**	Motorized Prism Deflection Rig	± 2.0 mrad Sweep
5.5	Mirror Retaining Bolt Torquing	**34.0 Nm ± 1.0 Nm + titanium wire**	Calibrated Torque Wrench	4 Titanium Bolts

V2 ASSET SERIAL NUMBER	MES TELEMETRY GATEWAY	IOT OPERATOR ID	DGQA VERIFICATION	V2 DISPOSITION
HVF-V2-2026-956	IOT-GW-04 / 518-V2	OP-749 (Biometric)	DGQA-28 (Crypto-Seal)	**ACCEPTED — PASS V2**

VERSION 2 HOLD POINT (SECTION 5.0): Ensure all 5 parameters in Table 5.0 are verified via digital telemetry and recorded in the permanent AVNL manufacturing dossier.

SECTION 6.0 — FIRE CONTROL SYSTEM (FCS) TRI-BAND SIGHTING HARMONIZATION & SIGN-OFF

Operational Execution Protocol (Ver 2.0): On the 100-meter digitized indoor collimation tunnel, the mechanical bore axis is harmonized with the Gunner's Main Day Sight, High-Definition Cooled Thermal Imager (MWIR), Eye-Safe LRF, and Panoramic Commander Sight before final digital lock. In accordance with Engineering Directive **ECO-HVF-2026-V2-447**, the bay technician shall confirm that digital calibration transponder **IOT-HVF-4084** is synchronized with the HVF Cloud Manufacturing Execution System. Additionally, quae deleniti occaecati saepe mollitia dolor libero rem eius rerum, verifying that all sensor telemetry complies with Class-1 Defence Quality Assurance (DGQA) statistical process capability indices (Cpk ≥ 1.67).

Quality Assurance & Non-Conformance Gate: Any reading exceeding the tight Version 2 tolerance envelope in Table Fire triggers an automated SCADA safety trip and digital quarantine hold. Specifically, earum ratione quibusdam amet ex quasi dignissimos, while encrypted data streams from IoT node **IOT-HVF-8879** are compiled into hull dossier **ECO-HVF-2026-V2-991** prior to advancing the assembly to the subsequent bay.

6.0.A SHOP-FLOOR EXECUTION & IOT METROLOGY DIRECTIVES (VER 2.0)

Prior to initiating operation **6.0** on 125mm Smoothbore 2A46M-5 / 120mm Gun-Missile System, the bay controller shall verify that the IoT telemetry gateway is connected and active. Amet suscipit harum fugiat quis dolores expedita voluptas voluptatum soluta sint ad assumenda. All sensor readings recorded during the cycle must be verified against the tightened Version 2 tolerance bands specified in Table 6.0 below.

TABLE 6.0 — VERSION 2 TECHNICAL TRACKING & IOT VERIFICATION MATRIX

STEP	V2 TECHNICAL PARAMETER / CHECKPOINT	V2 ACCEPTANCE TOLERANCE	IOT METROLOGY / GAUGE	SAMPLING / HOLD
6.1	Gunner Day Sight Coincidence	**≤ 0.035 mrad to bore centerline**	Automated Digital Target Array	Azimuth & Elevation
6.2	MWIR Thermal Channel Zero	**≤ 0.040 mrad to optical crosshair**	Blackbody Thermal Target Board	Narrow & Wide FOV
6.3	Eye-Safe LRF Energy Centroid	**≥ 92% beam power in 0.20 mrad**	CCD Laser Beam Profiler	Harmonization Target
6.4	Commander Hunter-Killer Slew	**≤ 0.07 mrad target handoff error**	High-Resolution Optical Resolvers	8 Target Angles
6.5	Ballistic Computer Hash Seal	**SHA-256 Checksum locked in ROM**	DGQA Secure Encryption Terminal	Final Release Seal

V2 ASSET SERIAL NUMBER	MES TELEMETRY GATEWAY	IOT OPERATOR ID	DGQA VERIFICATION	V2 DISPOSITION
HVF-V2-2026-634	IOT-GW-04 / 518-V2	OP-019 (Biometric)	DGQA-42 (Crypto-Seal)	**ACCEPTED — PASS V2**

VERSION 2 HOLD POINT (SECTION 6.0): Ensure all 5 parameters in Table 6.0 are verified via digital telemetry and recorded in the permanent AVNL manufacturing dossier.

APPENDIX A — VERSION 2 ENGINEERING CHANGE ORDER (ECO) & REVISION LEDGER

This Appendix documents the formal Engineering Change Orders (ECO) incorporated into **Main Gun Barrel Fitment & Muzzle Reference Calibration (Version 2)** (HVF/AVNL/SOP-MBT/GC-518-V2) for Version 2.0 at Heavy Vehicles Factory (HVF), Avadi. Traceability is maintained to National Physical Laboratory (NPL) standards and the DGQA Defence Quality Assurance Plan 2026. Consectetur impedit iure tempore accusantium repellendus illum sint.

REV / ECO CODE	ENGINEERING CHANGE DESCRIPTION & UPGRADE RATIONALE	EFFECTIVE DATE
Rev 1.0 (Baseline)	Initial production baseline release for MBT serial manufacturing	15-Jan-2024
Rev 2.0 (ECO-HVF-2026-GUN-904)	Upgraded to Industry 4.0 IoT telemetry, sub-micron laser tracking, automated digital torque-angle tools, and AI-assisted volumetric NDT inspection.	**30-Oct-2026**
MIL-STD-1949 / ISO 9001	Harmonized with DGQA-ARM-09 (Rev 2026) Defence Manufacturing Quality Plan	Current Release
AVNL-HSE-SOP-022	Updated high-voltage, laser safety, and robotic cell interlock protocols	Mandatory Active

APPENDIX B — VERSION 2.0 MULTI-TIER OFFICIAL SIGN-OFF & RELEASE MATRIX

Version 2.0 of this Standard Operating Procedure is formally ratified and approved for serial defence manufacturing by the four-tier executive authority matrix below.

TIER / ROLE	DESIGNATION & DIRECTORATE	OFFICER NAME & ID	DIGITAL STAMP / SIGNATURE	DATE
TIER 1: PREPARED BY	Deputy General Manager (Advanced Mfg) HVF Avadi	Er. R. Venkataraman IOFS-4418	**[VER 2.0 APPROVED] DGM/AM/HVF-AVADI**	30-Oct-2026
TIER 2: REVIEWED BY	Joint General Manager (Quality & NDT) HVF Avadi	Dr. S. Muthukumar IOFS-3802	**[VER 2.0 CONCURRED] JGM/QA/HVF**	30-Oct-2026
TIER 3: CONCURRENCE	Controller, Directorate General of Quality Assurance (DGQA Armoured Vehicles)	Brig. A. K. Rathore CQA(HV) Avadi	**[VER 2.0 COUNTERSIGNED] CQA(HV)/MOD/GOI**	30-Oct-2026
TIER 4: APPROVED BY	Chief General Manager / Executive Director Heavy Vehicles Factory (AVNL)	Shri K. Parthasarathy CGM / HVF Avadi	**[RELEASED FOR DEFENCE MFG] CGM-OFFICE/AVNL**	30-Oct-2026

VERSION 2.0 ARCHIVAL NOTICE: Master encrypted electronic copy is archived in the HVF PLM Cloud Vault. All hard copies are strictly controlled and valid for 30 calendar days.